

Fall 2019

Problem 1 (Fa19). Show that the limit $\lim_{R \rightarrow \infty} \int_0^R e^{ix^2} dx$ exists.

Problem 2 (Su84, Fa19). Show there is a unique continuous real valued function $f : [0, 1] \rightarrow \mathbb{R}$ such that

$$f(x) = \sin x + \int_0^1 \frac{f(y)}{e^{x+y}} dy$$

Problem 3 (Sp18, Fa19). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Show that

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) \sin nx \, dx = 0$$

Problem 4 (Su85, Fa19). Let $P(z)$ be a polynomial of degree $< k$ with complex coefficients. Let $\omega_1, \dots, \omega_k$ be the k -th roots of unity in $\mathbb{C}$. Prove that

$$\frac{1}{k} \sum_{i=1}^k P(\omega_i) = P(0)$$

Problem 5 (Fa86, Fa13, Fa19, Fa23). Show that the polynomial $p(z) = z^5 - 6z + 3$ has five distinct complex zeros, of which exactly three (and not five) are real.

Problem 6 (Fa79, Su81, Fa19). Let V be the vector space of sequences (a_n) of complex numbers. The shift operator $S : V \rightarrow V$ is defined by

$$S((a_1, a_2, a_3, \dots)) = (a_2, a_3, a_4, \dots)$$

1. Find the eigenvectors of S .
2. Show that the subspace W consisting of the sequences (x_n) with $x_{n+2} = x_{n+1} + x_n$ is a two-dimensional, S -invariant subspace of V and exhibit an explicit basis for W .
3. Find an explicit formula for the n -th Fibonacci number F_n , where $F_2 = F_1 = 1, F_{n+2} = F_{n+1} + F_n$ for $n \geq 1$.

Note: See also ??.

Problem 7 (Fa19). Let H_n , $n = 0, 1, 2, \dots$ be the basis in the space $\mathbb{R}[x]$ of real polynomials in one variable, obtained from the basis $1, x, x^2, \dots$ of monomials by the Gram–Schmidt orthogonalization procedure with respect to the inner product defined by $\langle u, v \rangle = \int_{-\infty}^{\infty} u(x)v(x)e^{-x^2/2} dx$. Show that H_n has n distinct real zeros.

Problem 8 (Su85, Fa19). By the Fundamental Theorem of Algebra, the polynomial $x^3 + 2x^2 + 7x + 1$ has three zeros, α_1 , α_2 , and α_3 . Compute $\alpha_1^3 + \alpha_2^3 + \alpha_3^3$.

Problem 9 (Fa19). Let $\mathbb{F}_2[x]$ be the ring of polynomials over the field with 2 elements.

1. For each degree 3 polynomial p in this ring, determine whether the quotient $\mathbb{F}_2[x]/\langle p \rangle$ by the principal ideal of p is a field.
2. For those p for which the quotients are fields, determine whether the fields are isomorphic to each other or not.
3. For those which are isomorphic, point out an isomorphism explicitly.

Problem 10 (Fa19). Let P_D denote the space of homogeneous degree D polynomials in $\mathbb{R}^n$. Prove that the kernel of the laplacian $\Delta : P_D \rightarrow P_{D-2}$, where $\Delta = \sum_{i=1}^n \partial^2 / \partial x_i^2$ is complementary to the image of the operator $r^2 P_{D-2} \rightarrow P_D$, defined as multiplication of a polynomial by $r^2 = \sum_{i=1}^n x_i^2$.

Problem 11 (Sp83, Sp01, Fa19). Suppose that f is a continuous function on $\mathbb{R}$ which is periodic with period 1. Show that:

1. The function f is bounded above and below and achieves its maximum and minimum.
2. The function f is uniformly continuous on $\mathbb{R}$.
3. There exists a real number x_0 such that $f(x_0 + \pi) = f(x_0)$.

Problem 12 (Fa19). Let p be a nonconstant polynomial with real coefficients such that all zeros are real and distinct. Show that for any real number r the polynomial $p + rp'$ has only real zeros.

Problem 13 (Sp79, Su85, Sp89, Fa19). Prove that if $1 < \lambda < \infty$, the function

$$f_\lambda(z) = z + \lambda - e^z$$

has only one zero in the half-plane $\Re z < 0$, and that this zero is real.

Problem 14 (Fa19). 1. Show that there is a function $g(z)$, holomorphic near 0, such that

$$g(z)^3 = e^{z^3 - z^7} - 1$$

for all z in a neighborhood of 0.

2. Find the radius of convergence of its power series about 0.

Problem 15 (Sp88, Fa19). Let X be a set and V a real vector space of real valued functions on X of dimension n , $0 < n < \infty$. Prove that there are n points $x_1, x_2, \dots, x_n$ in X such that the map $f \mapsto (f(x_1), \dots, f(x_n))$ of V to $\mathbb{R}^n$ is an isomorphism.

Problem 16 (Fa19). Let B be the 3×3 matrix whose diagonal entries are all 1 and whose other entries are all 2. Prove that there is no real matrix A with $B = AA^T$.

Problem 17 (Fa19). Let A and B be finite abelian p -groups isomorphic respectively to the direct sums of cyclic groups of orders $p^{a_1}, p^{a_2}, \dots, p^{a_k}, \dots$ where $a_1 \geq a_2 \geq \dots \geq a_k \geq \dots$, and $p^{b_1}, p^{b_2}, \dots, p^{b_k}, \dots$ where $b_1 \geq b_2 \geq \dots \geq b_k \geq \dots$. Prove that if A contains a subgroup isomorphic to B , then $a_k \geq b_k$ for all k .

Problem 18 (Fa19). Describe explicitly all nontrivial normal subgroups in the group G of rotations of the three-dimensional cube.